

Cheng Xin

Assistant Professor · California State University, Fresno, Department of Computer Science

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BIO

Cheng Xin is an Assistant Professor of Computer Science at **California State University, Fresno**. He received his Ph.D. in Computer Science from Purdue University under the supervision of Dr. Tamal K. Dey and was a postdoctoral researcher at Rutgers University advised by Dr. Jie Gao. His research develops **trustworthy and theoretically grounded AI systems** by connecting **topology, geometry, algorithms, and multi-agent interaction**. His work spans interpretable graph learning, non-Euclidean representations, topological data analysis, AI for Science, and reliable collective learning, with publications in **NeurIPS, ICML, CVPR, and SoCG**.

RESEARCH INTERESTS

- Topological Machine Learning
- Non-Euclidean Geometry in ML
- Multiparameter Persistence
- Interpretable AI
- Trustworthy & Multi-Agent AI
- AI for Science

ACADEMIC APPOINTMENTS & EDUCATION

California State University, Fresno Department of Computer Science

Assistant Professor of Computer Science

Aug 2026 – Present

- Research on **trustworthy and multi-agent AI**, topological and geometric learning, and theoretical foundations for reliable learning systems
- Teaching mathematical foundations of computer science, formal languages, and automata theory

Rutgers University Department of Computer Science

Postdoctoral Researcher, Advisor: Prof. Jie Gao

Oct 2023 – Aug 2026

- Developing topological frameworks for **interpretable AI**: TopInG achieves improved prediction accuracy and interpretability on molecular benchmarks
- Led research on **non-Euclidean representation learning**: Neuc-MDS and Johnson-Lindenstrauss extensions with provable theoretical guarantees
- Contributing to large-scale benchmarks for **3D/video generation**: DL3DV-10K dataset
- Designing algorithms for **multi-agent learning** in social settings

Purdue University Department of Computer Science

Ph.D. in Computer Science, Advisor: Prof. Tamal K. Dey

Aug 2020 – Aug 2023

- Dissertation: Decomposition and Stability of Multiparameter Persistence Modules

The Ohio State University Department of Computer Science and Engineering

Ph.D. in Computer Science, Advisor: Prof. Tamal K. Dey

Jan 2017 – Aug 2020

- Developed generalized persistence algorithms for multiparameter persistence modules

Lehigh University Department of Computer Science

M.S. in Computer Science, Advisor: Prof. Xiaolei Huang

Jan 2014 – May 2016

- Thesis: Machine Learning Techniques for Cervigram Image Analysis
- Research Focus: Medical image analysis, machine learning applications

Tongji University Shanghai, China

B.Eng. in Software Engineering

Sep 2009 – Jul 2013

INVITED TALKS

"TopInG: Topologically Interpretable Graph Learning via Persistent Rationale Filtration"
Conference on Topological Data Analysis: Recent Developments and Applications, University of Missouri, November 2025

"Understanding through Shape of Data: Topological Data Analysis for Interpretable AI"
Management Science and Information Systems Department Colloquium, Rutgers University, October 2024

"Exploring Representations Beyond Euclidean Geometry"
John Hopcroft Center Seminar, Shanghai Jiao Tong University, June 2024

"Generalized persistence algorithm for decomposing multi-parameter persistence modules"
Applied Algebraic Topology Network Seminar, July 2020

"Multiparameter Persistence and Its Applications"
Theory Seminar, Department of Computer Science, Rutgers University, November 2023

TEACHING EXPERIENCE

Instructor, Foundations of Computer Science and Introduction to Formal Languages and Automata, Fresno State, Fall 2026

Lecturer, Design and Analysis of Algorithms (graduate course, 45 students), 2025 Fall

Teaching Assistant, Data Structures and Algorithms (undergraduate, 200 students), 2023 Spring

Teaching Assistant, Computational Geometry (graduate, 30 students), 2020 Fall

PROFESSIONAL SERVICE

Area Chair, TAG-DS Workshop, 2026

Reviewer, ICML, ICLR, NeurIPS, SoCG

PUBLICATIONS

Machine Learning (NeurIPS, ICML, CVPR, TMLR)

- ICML 2025** C. Xin, F. Xu, X. Ding, J. Gao, J. Ding. "TopInG: Topologically Interpretable Learning via Persistent Rationale Filtration"
- NeurIPS 2025** C. Deng, J. Gao, K. Lu, F. Luo, C. Xin[†]. "Johnson-Lindenstrauss Lemma Beyond Euclidean Geometry"
- NeurIPS 2024** C. Deng, J. Gao, K. Lu, F. Luo, H. Sun, C. Xin[†]. "Neuc-MDS: Non-Euclidean Multidimensional Scaling Through Bilinear Forms"
- ICML 2024** S. Haddadan, C. Xin, J. Gao. "Optimally Improving Cooperative Learning in a Social Setting"
- CVPR 2024** L. Ling, ..., C. Xin, et al. "DL3DV-10K: A Large-Scale Scene Dataset for Deep Learning-Based 3D Vision"
- TMLR 2024** S. Zhang, C. Xin, T. K. Dey. "Expressive Higher-Order Link Prediction through Hypergraph Symmetry Breaking"
- ICML-W 2023** C. Xin, S. Mukherjee, S. N. Samaga, T. K. Dey. "GRIL: A 2-parameter Persistence Based Vectorization for Machine Learning"

Computational Geometry & Topology (SoCG, JACT)

- SoCG 2026** S. Mukherjee, S. N. Samaga, C. Xin, S. Oudot, T. K. Dey. "D-GRIL: End-to-End Topological Learning with 2-parameter Persistence"
- SoCG 2026** C. Deng, J. Gao, K. Lu, F. Luo, C. Xin. "Locality Sensitive Hashing in Hyperbolic Space"
- JACT 2022** T. K. Dey, C. Xin[†]. "Generalized persistence algorithm for decomposing multiparameter persistence modules"
- arXiv 2021** T. K. Dey, C. Xin[†]. "Rectangular Approximation and Stability of 2-parameter Persistence Modules"
- SoCG 2018** T. K. Dey, C. Xin[†]. "Computing Bottleneck Distance for 2-D Interval Decomposable Modules"

Medical Imaging & Pattern Recognition

- PR 2017** T. Xu, H. Zhang, C. Xin, et al. "Multi-feature based benchmark for cervical dysplasia classification evaluation"
- MLMI 2015** T. Xu, C. Xin^{*} et al. "A New Image Data Set and Benchmark for Cervical Dysplasia Classification Evaluation"

[†] authors alphabetically ordered · ^{*} co-first author

INDUSTRIAL EXPERIENCE

Electronic Arts (EA) Machine Learning Scientist Intern May – Aug 2018
Big Data Group · Redwood City, CA

- Large-scale machine learning on **Spark**
- Graph learning on relational database, attributes evaluation and selection, dataset compression

Amazon Software Development Engineer Intern May – Aug 2015
AWS Infrastructure Group · Seattle, WA

- Data management system for network messages supporting receiving, parsing, storing, and retrieving

SKILLS

Python PyTorch Spark Keras Java C C++ MATLAB R

REFERENCES

Confidential recommendation letters available via Interfolio Dossier Delivery.

Dr. Tamal K. Dey Professor, Computer Science, Purdue University tamaldey@purdue.edu	Dr. Jie Gao Professor, Computer Science, Rutgers University jg1555@cs.rutgers.edu	Dr. Feng Luo Professor, Mathematics, Rutgers University fluo@math.rutgers.edu
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